ZIYANG CHEN

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Research Interests

Multimodal learning across **vision**, **audio** and **language**; unified multimodal foundation models; diffusion models for multimodal generation.

Work Experience

Research Lead, Luma AI, CA Jan. 2026 – Present

Pre-training: Led audio pretraining and joint audio-video generation across modeling, data, and infrastructure.

Research Scientist, Luma AI, CA June 2025 – Dec. 2025

Post-training: Enabled audio-conditioned generation and distilled video models for efficient inference.

Research Scientist Intern, Adobe Research, CA May 2024 – Nov. 2024

Audio generation: Controllable video-guided Foley generation with diffusion models.

Research Scientist Intern, Meta Reality Labs, PA May 2023 – Nov. 2023

Spatial audio: Real-world acoustic modeling from audio-visual data.

Selected Projects

Joint Audio-Video Generation Ray3 & Uni-1 Luma AI

- Developed an in-house audio variational autoencoder for diffusion-based generation.
- Designed and implemented a unified training framework for multimodal generation.
- Led audio pretraining across speech, sound effects, and music for the Ray3 and Uni model families.
- Designed and implemented joint audio-video pretraining for synchronized audiovisual generation in Ray3 and Uni-1.
- Led the development of large-scale audio and audio-video data curation and annotation pipelines.

Video Avatar Real-Time Speech-to-Video Model Luma AI

- Designed the model architecture for speech-to-video generation and built the training, inference, and data-filtering pipelines from scratch.
- Applied diffusion distillation to enable avatar interaction driven by a background LLM agent in real-time.

Ray3.14 Flagship Few-Step Video Generation Model Luma AI

- Core contributor to distillation pipeline and algorithms, enabling fast few-step generation across diverse downstream tasks.

Education

University of Michigan, Ann Arbor Jan. 2022 – May 2025

Ph.D. in Electrical & Computer Engineering | Advisor: Prof. *Andrew Owens*

University of Michigan, Ann Arbor Sep. 2019 – May 2021

M.S.E. in Computer Vision | Advisor: Prof. *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sep. 2015 – Aug. 2019

B.S.E. in Electrical & Computer Engineering | Minor in Entrepreneurship

Graduated *magna cum laude*

Publications

* indicates equal contribution

arXiv 2026 **Luma Uni-1: A Frontier Unified Understanding and Generation Model** Blog

Luma AI · Coming soon

CVPR 2025 **Video-Guided Foley Sound Generation with Multimodal Controls** Project site

Ziyang Chen, Prem Seetharaman, Bryan Russell, Oriol Nieto, David Bourgin, Andrew Owens and Justin Salamon.

CVPR 2025 **GPS as a Control Signal for Image Generation** Project site
Chao Feng, Ziyang Chen, Aleksander Holynski, Alexei A. Efros, and Andrew Owens.

CVPR 2025 **Supervising Sound Localization using In-the-wild Egomotion**
Highlight
Anna Min, Ziyang Chen, Hang Zhao, and Andrew Owens.

NeurIPS 2024 **Images that Sound: Composing Images and Sounds on a Single Canvas** Project site
Ziyang Chen, Daniel Geng and Andrew Owens.

CVPR 2024 **Real Acoustic Fields: An Audio-Visual Room Acoustics Dataset and Benchmark** Project site
Highlight
Ziyang Chen, Israel D. Gebru, Christian Richardt, Anurag Kumar, William Laney, Andrew Owens, and Alexander Richard.

CVPR 2024 **Binding Touch to Everything: Learning Unified Multimodal Tactile Representations**
Project site
Fengyu Yang*, Chao Feng*, Ziyang Chen*, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong.

ICCV 2023 **Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation**
Project site
Ziyang Chen, Shengyi Qian and Andrew Owens.

CVPR 2023 **Conditional Generation of Audio from Video via Foley Analogies** Project site
Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.

CVPR 2023 **Self-Supervised Video Forensics by Audio-Visual Anomaly Detection** Project site
Highlight
Chao Feng, Ziyang Chen and Andrew Owens.

ECCV 2022 **Sound Localization by Self-Supervised Time Delay Estimation** Project site
Ziyang Chen, David F. Fouhey and Andrew Owens.

CVPR 2022 **Mix and Localize: Localizing Sound Sources in Mixtures** Project site
Xixi Hu*, Ziyang Chen* and Andrew Owens.

CoRL 2021 **Structure from Silence: Learning Scene Structure from Ambient Sound** Project site
Oral
Ziyang Chen*, Xixi Hu* and Andrew Owens.

Academic Service

Workshop Organizer
Generative AI for Audio-Visual Content Creation ICCV 2025 · ECCV 2026
Sight and Sound CVPR 2024 · CVPR 2025 · CVPR 2026

Reviewer
Conferences CVPR (2023–2026), ICCV (2023, 2025), ECCV (2024), NeurIPS (2024), SIGGRAPH (2024), WACV (2023)
Journals IJCV

Awards & Scholarships

Lums Fellowship, University of Michigan, Ann Arbor 2019–2021
Outstanding Graduate — Class of 2019, Shanghai Jiao Tong University 2019
Academic Excellence Scholarship, Shanghai Jiao Tong University 2016–2018
Dean's List, Shanghai Jiao Tong University, SJTU–UM JI 2015–2018